

Tensor Analysis

8. Line Element and Metric Tensor

The differential of arc length ds in rectangular coordinates (x, y, z) is obtained from $ds^2 = dx^2 + dy^2 + dz^2$. By transforming to general curvilinear coordinates, this becomes

$$ds^2 = \sum_{p=1}^3 \sum_{q=1}^3 g_{pq} du_p du_q$$

Such spaces are called three-dimensional Euclidean spaces.

A generalization to N -dimensional space with coordinates $(x^1, x^2, \dots, x^N)$ is immediate. We define the line element ds in this space to be given by the quadratic form, called the metric form or metric.

$$ds^2 = \sum_{j=1}^N \sum_{k=1}^N g_{jk} dx^j dx^k$$

or, using the summation convention,

$$ds^2 = g_{\rho} dx^{\rho} dx^{\rho}$$

In the special case where there exists a transformation of coordinates from x^j to x^k such that the metric form is transformed into $(d\bar{x}^1)^2 + (d\bar{x}^2)^2 + \dots + (d\bar{x}^N)^2$ or $d\bar{x}^i d\bar{x}^i$, then the space is called N-dimensional Euclidean space. In the general case, however, the space is called Riemannian.

The quantities $g_{\rho\sigma}$ are the components of a covariant tensor of rank two called the metric tensor or fundamental tensor. We can, and always will, choose this tensor to be symmetric Conjugate or Reciprocal Tensors.

Let $g = |g_{\rho\sigma}|$ denote the determinant with elements $g_{\rho\sigma}$ and suppose $g \neq 0$. Define $g^{\rho\sigma}$ by

$$g^{\rho\sigma} = \frac{\text{cofactor of } g_{\sigma\rho}}{g}$$

Then $g^{\rho\sigma}$ is a symmetric contravariant tensor of rank two called the conjugate or reciprocal tensor. It can be shown that

$$g^{\rho\sigma} g_{\sigma\tau} = \delta^{\rho}_{\tau}$$